
MIAC BASKETBALL SCHEDULE PROJECT (18 Conference Games)

— Alex McKeever —

Creation Process

The Plan

- 18 Game Conference Schedule
 - Men have 8 repeated opponents (2 are not repeated)
 - Women have 7 repeated opponents (4 are not repeated)
- Saturday (Even Rounds) Double Headers
- Midweek (Odd Rounds) Independent
 - That is, we do not force double headers but they could happen
- Similar methodology of generating many schedules and filtering to the best one(s)

Finding The Repeat Opponents

- Goal is to protect CON, CSS, SMU
 - GAC, SJU to some extent
- Pick opponents based on distance metrics as before
- Some element of randomness for the not obvious choices (i.e. city teams)
- Code is easily adaptable if we want to make adjustments
- We will analyze distance metrics of many generated schedules to ensure fairness

M	AUG	BET	CAR	CON	GAC	HAM	MAC	SJU	SMU	STO	CSS
AUG		18	50	212	72	12	13	77	126	47	142
BET	18		59	209	82	16	20	74	131	56	134
CAR	50	59		251	61	53	50	116	102	7	179
CON	212	209	251		258	217	218	142	330	248	255
GAC	72	82	61	258		79	77	123	137	59	206
HAM	12	16	53	217	79		11	82	124	49	141
MAC	13	20	50	218	77	11		84	125	47	143
SJU	77	74	116	142	123	82	84		195	113	167
SMU	126	131	102	330	137	124	125	195		105	229
STO	47	56	7	248	59	49	47	113	105		177
CSS	142	134	179	255	206	141	143	167	229	177	

Schedule Saturdays

- We simply pick 9 (unique!) match-ups and schedule them in the even rounds
- Moreover, we take the last 2 match-ups and make them double headers in rounds 1 and 3 (for 11 total base games)
 - This gets a Women's round robin
 - This gets a Men's round robin with one bye
- This method ensures that each Men's team has one bye week so far
- We also restrict so that each team has even home/away split

Schedule Remaining Weeks

- For men, we simply schedule the previously chosen repeat games and flip home/away location
 - Since one Men's team *needs* a bye each round (odd number of teams) we schedule the 17th and 18th conference games in a pseudo week 19 and 20
 - Some teams will only have one game to fit in while others will have two
 - These could be played whenever of course and not necessarily after week 18
 - Perhaps an extra midweek game somewhere?
- For women, we simply schedule the previously chosen repeat games **INDEPENDENTLY** and flip home/away location
 - No major issues here since we have no bye weeks to worry about

Schedule Checks

- We ensure that every single team has 9 home and 9 away games
- Every Men's team gets two byes (18 games total) when considering the extra two rounds that Men's team must play
 - Due to byes resulting from an odd number of teams
- Every once in a while a DH will get split to ensure the 9/9 home/away split but since we added the extra two weeks of DH every team will still have 9 double headers!
 - (This could be tinkered with as well)
- Games are also scheduled in a way such that there are (almost always) at least three round between repeat match-ups

Recap

- 18 Weeks
- DH are the repeat matchups
- Mid-week games are base opponents
 - Men's teams will have 1 or 2 extra base games they need to play
 - Though this allows for some flexibility in the schedule
- Metrics
 - Team pain and home/away streaks will be more of a focus since distance should be less variable since we get most repeat opponents
 - Instead of just picking half to play again, we pick all but 2/4
 - No issues with ensuring con/non-con distance fairness

Filter Metrics!

Filter

Based on our schedule creation, what do we have left to filter?

- Ensure each team gets a home game in one of week 16, 17, or 18
- Analyze home/away streaks to ensure fairness
- Analyze distances between Men's bye weeks for fairness
- Distance checks to ensure fairness
 - These are much more minimal than the first iteration of this project (10 Con, 6 Non-Con) due to more games being scheduled and them all being conference games

Filter Metrics Explained

- **RMSE_Total_Dist**
 - Captures the fairness of distance traveled across all games (lower is more fair)
- **RMSE_TeamPain**
 - Captures the overall fairness of the schedule (lower is more fair)
- **AvgPainFromMean**
 - Captures the overall ease of the schedule (lower is easier schedule)
- **AvgDistFromMean**
 - Captures the overall distance traveled (lower is less distance)

The RMSE metrics capture variability. The lower they are, the more equitable the schedule is.

To calculate RMSE we first take the average of the metric across all teams. We then calculate the difference between the team and average and square it. We sum these squared values and then take the square root. The square/square root process helps to flag when island schools have hard schedules.

The average from mean metrics capture overall “goodness”. **The lower they are, the more we are minimizing distance (cost) or bad schedule instances like long home/away streaks.**

To calculate average from mean we first take the average of the metric across all generated schedules for each team. We then calculate the difference between the metric and the average for each team. We then take the averages of these differences across all 11 teams.

On the whole, these metrics account for schedule fairness and goodness relative to geographical limitations (since some teams will **ALWAYS have worse schedules based on geography. DHs and our repeat opponent selection methodology help make schedules as fair as possible however.**

A Note on Distances

Based on the nature of the schedule construction, **distances do not vary by much in generated schedules.** This is again because every team plays every other team at both home and away locales except for 2/4 other teams. Combined with our logic for removing the furthest repeat opponents for geographically far teams, there is not many different distances possible. In other words, we have mostly removed the bad distances! **As such, we focus more on Team Pain metrics when selecting schedules.**

Proposed Schedules!

MEN

Bye

Away

Home

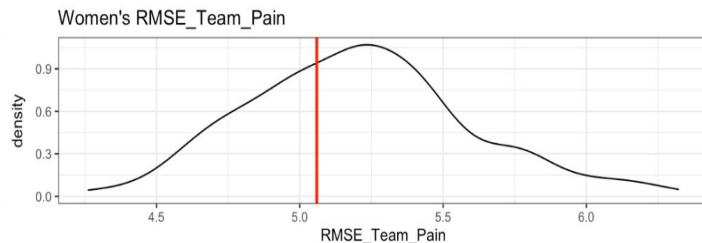
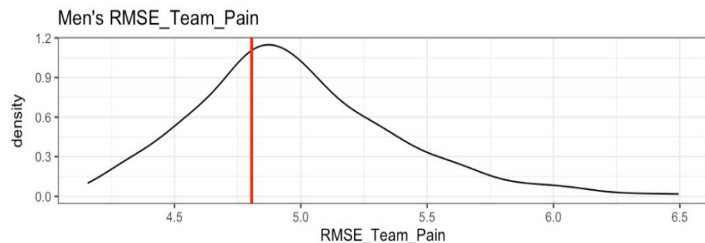
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WOMEN

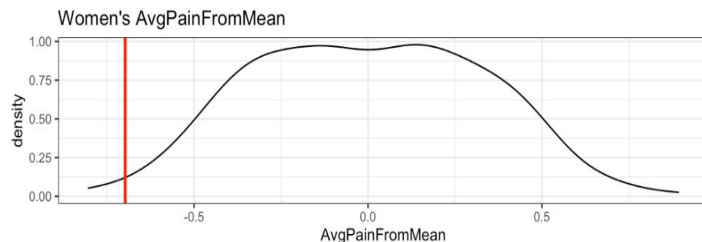
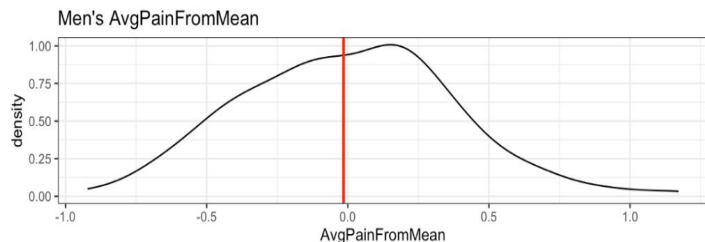
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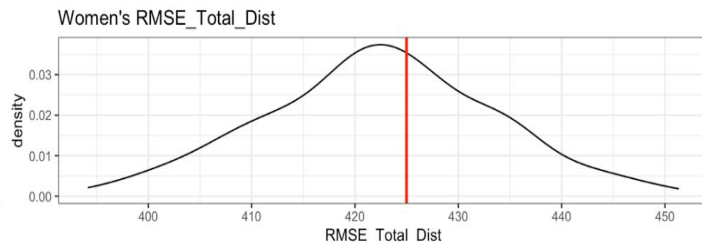
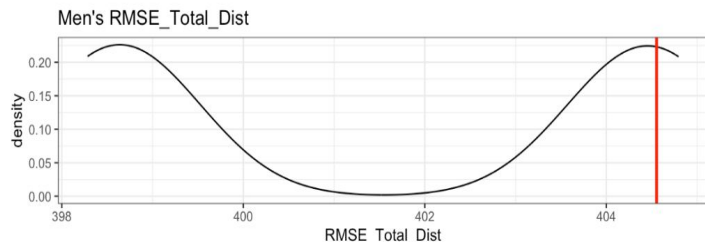
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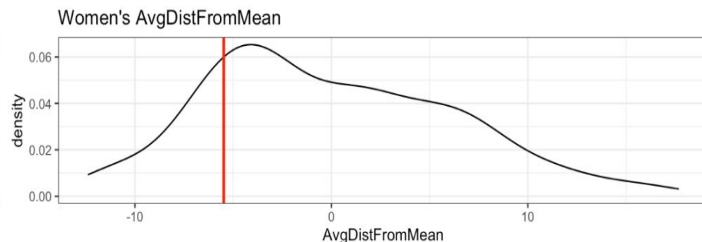
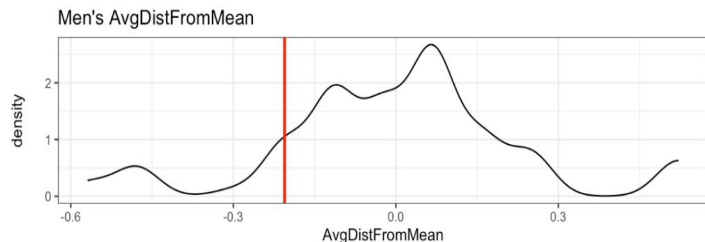
**Are Team Pain Metrics Fair?
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**Are Team Pain Metrics Small Compared to Other Schedules?
(Lower = Better)**



**Are Distance Metrics Fair?
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The Bottom 4 Plots Mean Less Since Distance is Always Pretty Good

MEN

Bye

Away

Home



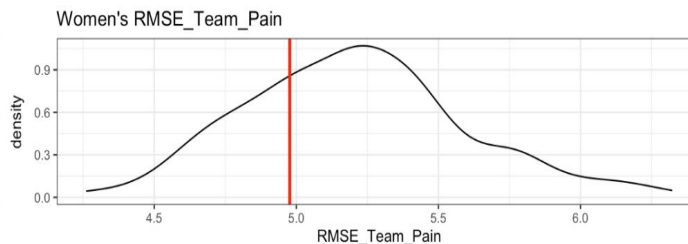
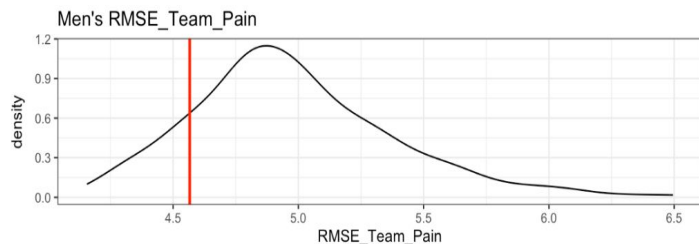
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WOMEN

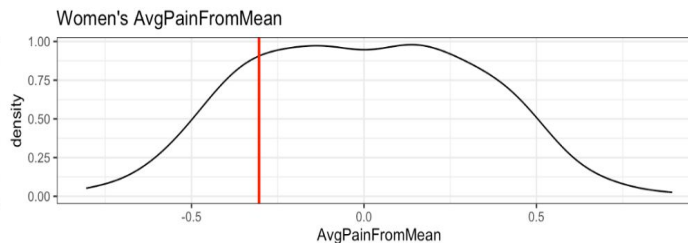
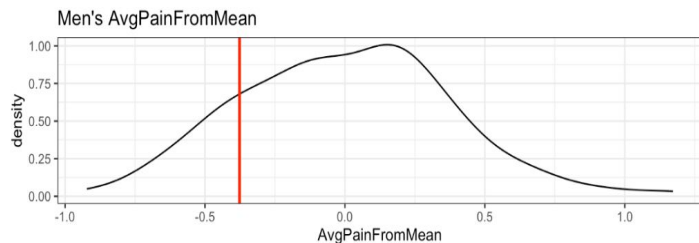
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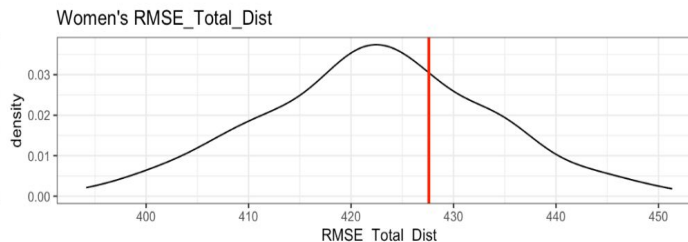
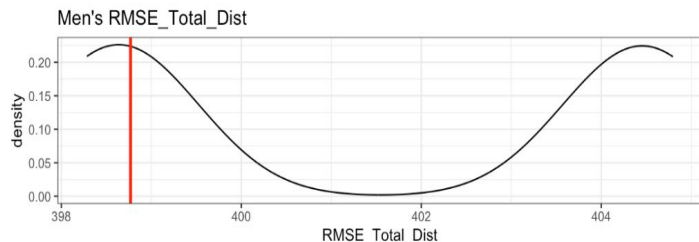
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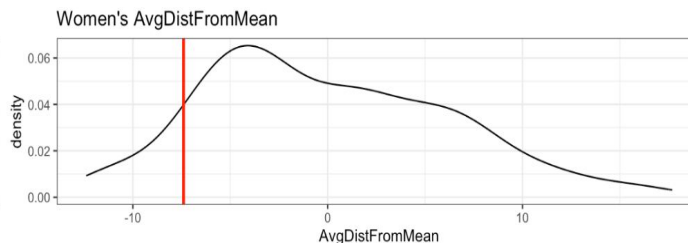
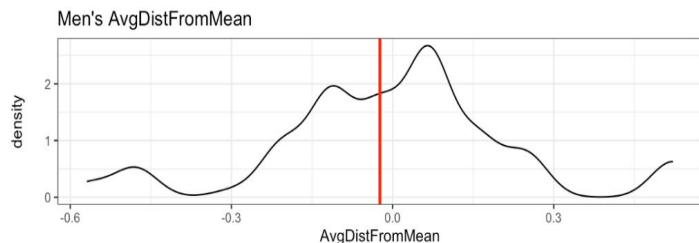
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MEN

Bye

Away

Home

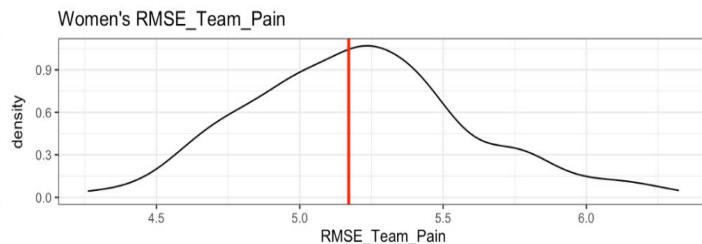
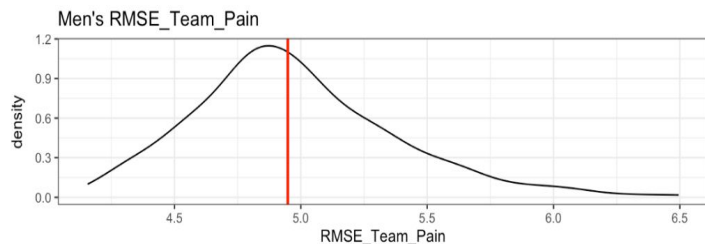
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WOMEN

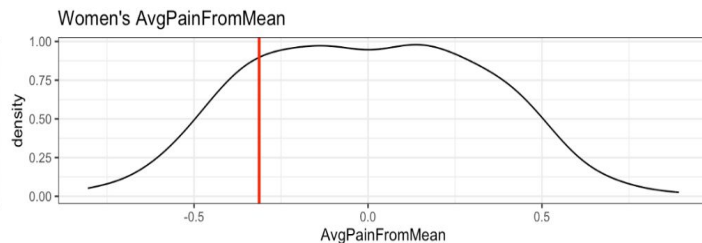
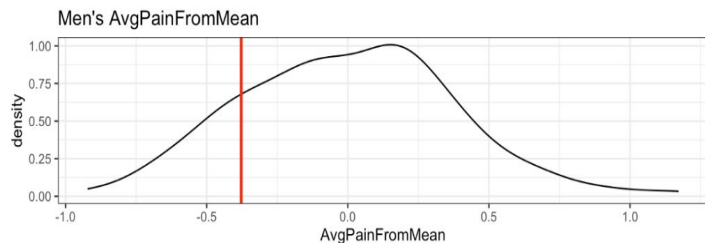
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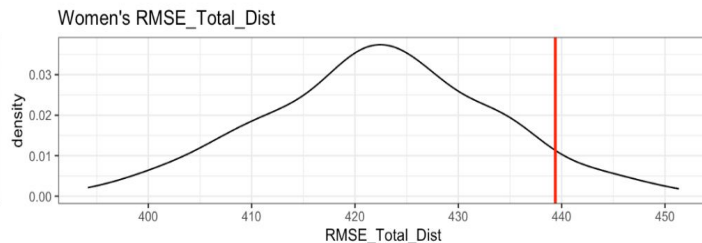
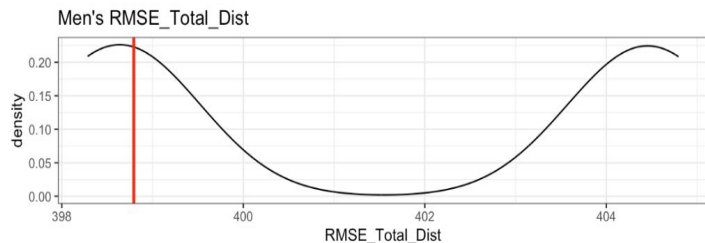
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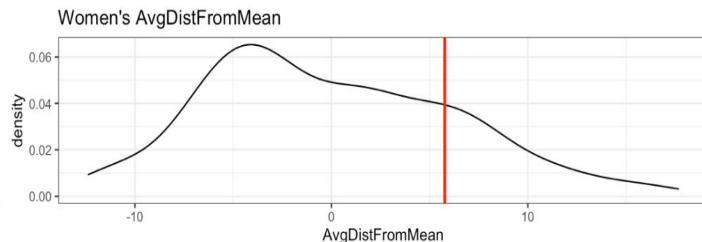
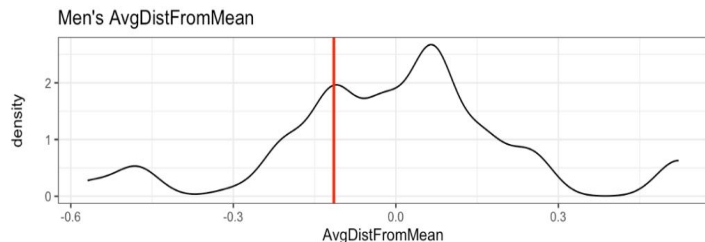
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